EXPERIMENT2

AIM: To convert an ER Diagram into a relational schema and implement it using MYSQL.

Relational SCHEMA:

CUSTOMER (Customer_ID, NAME, Email, Phone)
PRODUCE (PRODUCT_ID, Product_Name, Price, category_ID)
CATEGORY (Category_ID, category_Name)
ORDERS (ORDER_ID, ORDER_DATE, CUSTOMER_ID)
ORDER_ITEM (ORDER_ID, Product_ID, Quantity Price)

SQL:

```
CREATE TABLE CUSTOMER(
CUSTOMER_ID INT PRIMARY KEY,
NAME VARCHAR(50) NOT NULL,
EMAIL VARCHAR(100) UNIQUE);

CREATE TABLE CATEGORY(
CATEGORY_ID INT PRIMARY KEY,
CATEGORY_NAME VARCHAR(50));

CREATE TABLE PRODUCT(
PRODUCT_ID INT PRIMARY KEY,
PRODUCT_NAME VARCHAR(50),
PRICE DECIMAL(10.2),
CATEGORY_ID INT,
FOREIGN KEY(Category_ID)
REFERENCES CATEGORY(Category_ID));

CREATE TABLE ORDERS(
Order_ID INT PRIMARY KEY,
Order_Date DATE
Customer_ID INT,
FOREIGN KEY (CUSTOMER_ID)
REFERENCES CUSTOMER(CUSTOMER_ID));
```

```
CREATE TABLE ORDER_ITEMC
ORDER_ID INT,
Product_ID INT,
Quantity INT,
Price DECIMAL(10,2),
PRIMARY KEY(Order_ID, Product_ID),
FOREIGN KEY(Order_ID)
REFERENCES ORDERS(ORDER_ID)
 ON DELETE CASCADE);
```

SAMPLE DATA:

```
INSERT INTO CUSTOMER VALUES
(1,'RAHUL', rahul@gmail.com);

INSERT INTO CATEGORY VALUES
(101,'ELECTRONICS');

INSERT INTO PRODUCT VALUES
(201,'LAPTOP', 65000, 101);
```

VIOLATION:

```
INSERT INTO ORDERS VALUES
(501,'2026-08-19',99);
```

esult: foreign key error because customer_ID 99 does not exist.

CONCLUSION:

The ER diagram was successfully converted into a relational schema with primary keys, foreign keys NOT, NULL, UNIQUE, and ON DELETE CASCADE CONSTRAINTS.